Retrofit Radar: Project Report

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1. Essay: which buildings should we fix first?

Buildings account for roughly a third of the world's energy use and energy-related CO_2 emissions. Retrofits such as insulation, heat pumps and HVAC upgrades reduce that, but they are expensive. An organisation with hundreds of buildings cannot fix them all at once, so it has to choose. Ranking buildings by raw energy use does not help, because a hospital always uses more than a warehouse and a building in a cold state always heats more. The useful question is whether a building uses more than comparable buildings in the same climate.

Retrofit Radar answers that question in MATLAB. Using the WiDS Datathon 2022 data (75,757 building-years with building characteristics and weather), we predict each building's site energy use intensity (EUI), the energy it uses per square foot per year. We then turn the prediction into a decision tool. A benchmark model estimates what each building should use from its type, size, age, state and weather. Buildings using at least 1.5 times their expected energy become retrofit candidates, ranked by wasted energy and CO_2.

We treated data handling as the core of the project. The real test set is a future year (Year 7) with no buildings from State 6, which makes up 67% of the training rows. The same building also reappears across years: we built a pseudo building ID and measured that the model predicts buildings it has seen before far better (error 30) than new ones (error 46). So we never split randomly. We tuned only on Years 1–5 and scored every model on Year 6, which no decision ever touched. Missing values were filled using statistics from training rows only. When tuning on a single year backfired, we diagnosed why and switched to time-series cross-validation.

We compared eight models, from a simple baseline through linear regression, decision trees, bagging, gradient boosting and a neural network. The final gradient-boosted model reaches an RMSE of 40.3 and an R^2 of 0.50 on Year 6, 20% better than the baseline. Modelling the log of energy use was the largest single improvement. Our MATLAB app then showed that the best model depends on the segment. Outside State 6, a raw-target boosted model does better (RMSE 40.2 vs 43.2). We report that openly rather than swapping models after seeing the test year.

The retrofit engine flags 2,973 of 22,444 Year 6 buildings (13%). It sets aside 107 implausible readings for meter checks instead of counting them as savings. The benchmark never sees Energy Star ratings, yet the buildings it flags have a median rating of 20, against 73 for the rest. Two independent measures of efficiency agree. The waste is concentrated: the top 5% of buildings hold 57% of all excess energy. If every candidate matched its benchmark, the saving would be up to about 8,200 GWh and 3.1 million tonnes of CO_2 a year, at an assumed emission factor.

Retrofit Radar is a screening tool, not an engineering audit. A flag says which building to inspect, not what to fix. The next steps are adding retrofit costs to rank buildings by cost per tonne saved, and choosing models per building type.

2. Results at a glance

All accuracy figures are measured on Year 6 (22,444 buildings), a year never used for any modelling decision.

Measure	Result
Final model	Gradient-boosted trees (LSBoost) on log(site EUI): 298 trees, up to 60 splits each, learning rate 0.05
RMSE / MAE / R^2	40.3 / 18.7 kBtu/ft²/yr / 0.50
Improvement over the facility-type baseline	20% lower RMSE (50.7 → 40.3)
RMSE excluding State 6 (closer to the test year)	43.2 (best model outside State 6: raw-target boosting, 40.2)
Memorisation check	RMSE 29.7 on buildings seen in earlier years vs 45.6 on new buildings
Retrofit candidates (1.5–5× expected energy)	2,973 of 22,444 buildings (13.2%)
Readings to verify (≥ 5× expected)	107 buildings
Independent validation	Median Energy Star rating: candidates 20, unflagged buildings 73
Concentration of waste	Top 5% of buildings hold 57% of all excess energy
Excess energy of candidates (upper bound)	About 8,215 GWh/yr; about 3,122 kt CO_2/yr at an assumed 0.38 t/MWh

Measure	Result
Median excess per candidate	About 1,305 MWh/yr

3. The data

We used the WiDS Datathon 2022 building-energy dataset (Climate Change AI and MIT, via Kaggle), as supplied by the organisers. One row is one building in one year. Years and states are anonymised: we know their order and grouping but not the real calendar years or state names.

File	Rows	Columns	Years	Target included?	How we used it
`train.csv`	75,757	64	1–6	Yes (`site_eui`)	All training, tuning and testing
`test.csv`	9,705	63	7	No	Final predictions only (cannot be scored)

Rows per year in `train.csv` : Year 1: 2,117 · Year 2: 9,058 · Year 3: 10,879 · Year 4: 12,946 · Year 5: 18,308 · Year 6: 22,449.

Every column group, and what we did with it:

Group	Columns	Missing in train / test	Used?
Identifiers	`id` (row ID), `Year_Factor` (anonymised year 1–7)	0%	`id` for output only; `Year_Factor` for splitting, not as an input
Location	`State_Factor` (7 anonymised states), `ELEVATION`	0%	Yes
Building	`building_class` (Residential / Commercial), `facility_type` (60 types), `floor_area` (ft²), `year_built`	year built: 2.4% / 0.9%	Yes (floor area as its log)
Efficiency	`energy_star_rating` (1–100)	35.3% / 23.2%	Accuracy model only, plus a missing flag
Monthly temperatures	Minimum, average and maximum for each of 12 months (36 columns)	0%	Summarised into winter minimum and summer maximum

Group	Columns	Missing in train / test	Used?
Annual climate	Cooling and heating degree days, precipitation, snowfall, snow depth, average temperature	0%	Yes
Temperature extremes	Days below 30, 20, 10, 0 °F; days above 80, 90, 100, 110 °F	0%	Yes
Wind and fog	Max wind speed, its direction, peak wind direction, days with fog	54–60% / 88–94%	Dropped
Target	`site_eui`: site energy use intensity, kBtu/ft²/yr	0% (train only)	Predicted (as its log)

What exploring the data showed:

1. The target is heavily right-skewed: median 75, mean 83, maximum 998 kBtu/ft²/yr. On a log scale it is close to symmetric (figure 1).
2. Four wind and fog columns are mostly empty, and emptier still in the test year (figure 2).
3. State 6 is 67% of training rows but 0% of test rows (figure 3), a strong distribution shift.
4. The same building reappears in several years: 75,745 rows (after removing duplicates) come from 56,382 distinct buildings.
5. Small quality problems: 12 exact duplicate rows and 25 impossible build years (such as 0).

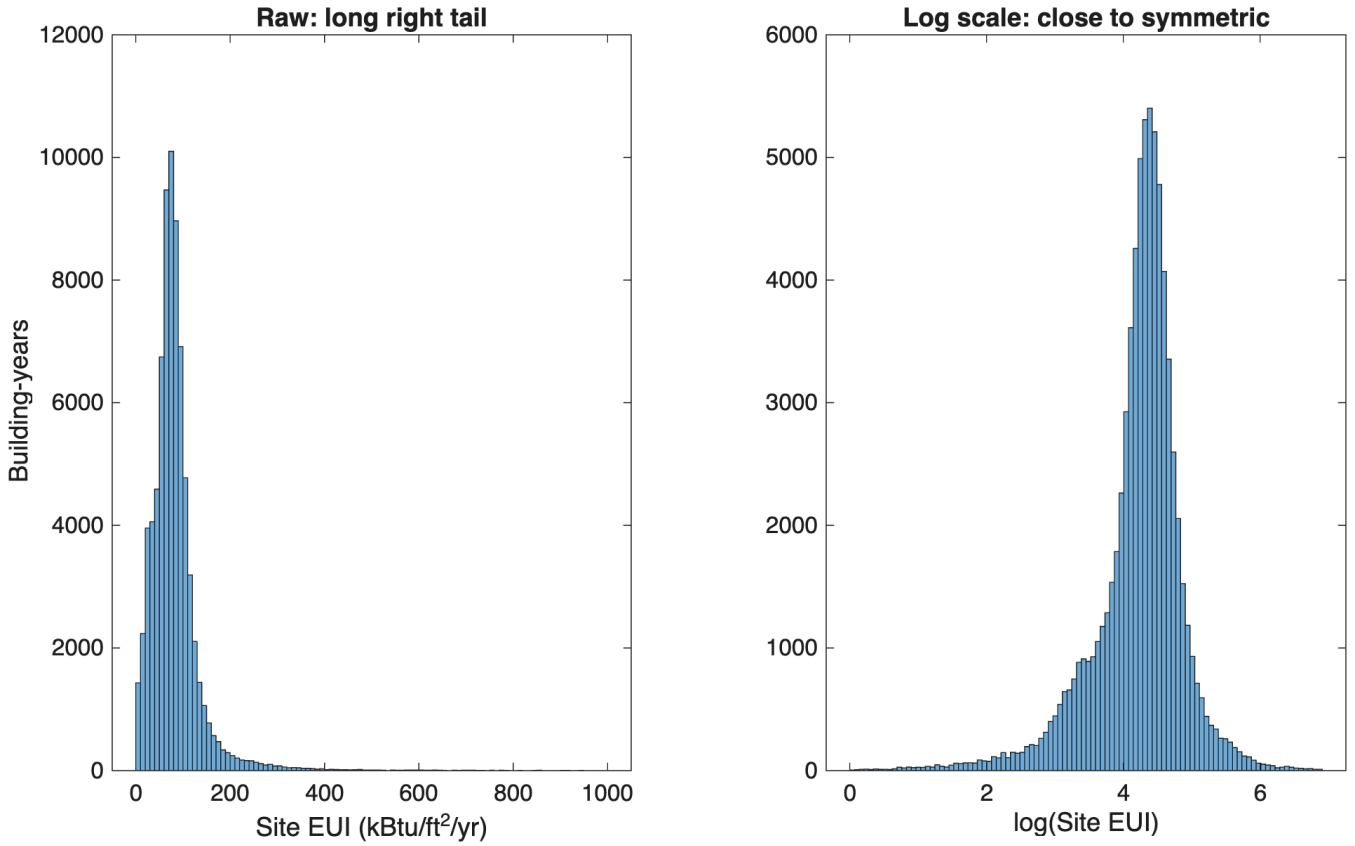


Figure 1. Site EUI is heavily right-skewed (left); on a log scale it is close to symmetric (right), which is why we model log(EUI).

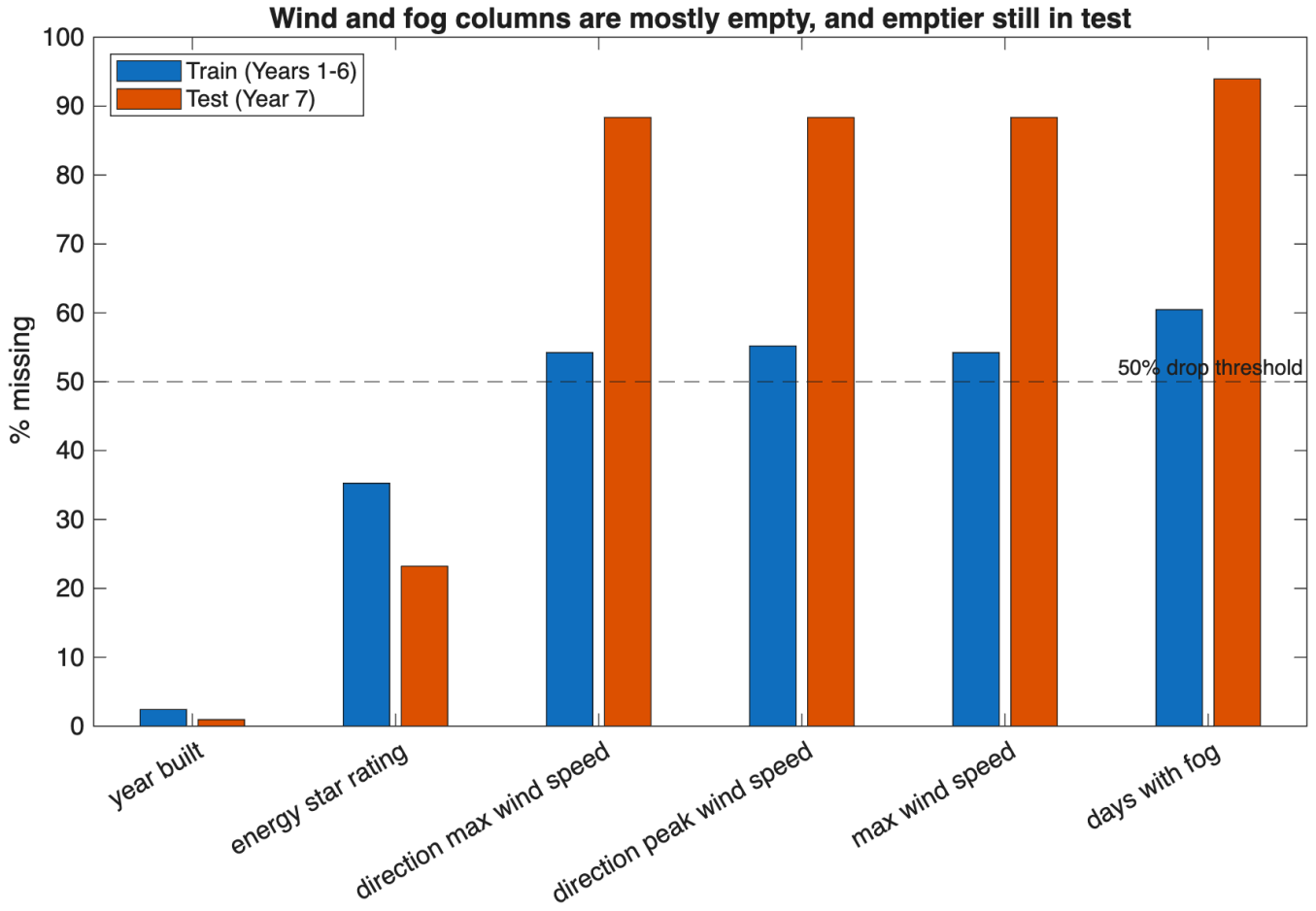


Figure 2. Missing values per column. The wind and fog columns cross the 50% threshold and are dropped; Energy Star and year built are kept and filled.

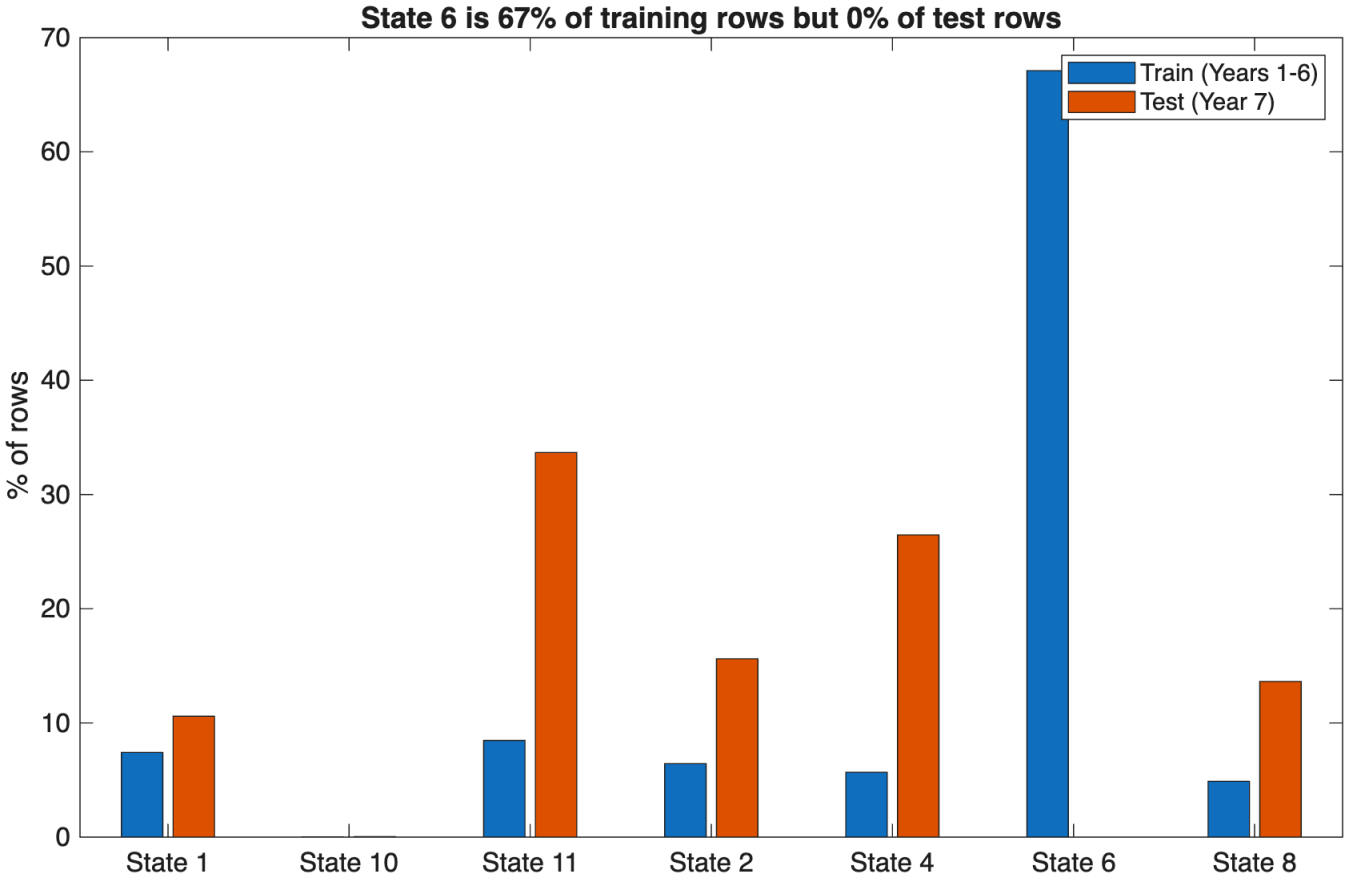


Figure 3. State 6 dominates training (67%) but is absent from the test year, so we also report accuracy excluding State 6.

4. The process, step by step

The whole pipeline is one MATLAB script, `BuildingEnergyRetrofit.m`, run from `main.m`. It executes these steps in order.

Step 1: Load. Both CSVs are read with `detectImportOptions`; state, building class and facility type are imported as `categorical` (labels, not numbers). The test file is given the same category lists as training with `setcats`.

Step 2: Clean.

Action	Rows or columns affected	Reason
Drop columns more than 50% empty	4 wind and fog columns	Filling them would mostly invent data

Action	Rows or columns affected	Reason
Remove exact duplicate rows	12 rows	Duplicates over-weight buildings and can fall on both sides of a split
Set `year_built` below 1800 to missing	25 rows	Values such as 0 are data-entry errors
Fill missing `energy_star_rating` and `year_built` with the median for that facility type	35% and 2% of rows	A hospital's missing value is better estimated from other hospitals
Compute those medians from the training rows only	Every split	Using test rows would leak test information into training

Step 3: Engineer features.

New feature	Definition	Reason
`log_floor_area`	log(floor area)	Floor area spans about 1,000 to 6.4 million ft²
`winter_min_temp`	Mean of December, January and February minimum temperatures	Replaces 36 near-duplicate monthly columns (figure 4)
`summer_max_temp`	Mean of June, July and August maximum temperatures	As above
`star_missing`	1 if the Energy Star rating was missing, else 0	Having no rating at all carries information
`building_id` (not an input)	Group of rows with the same state, facility type, floor area, year built and elevation	Tracks the same building across years

The final accuracy model uses 24 inputs: state, building class, facility type, log floor area, year built, Energy Star rating, `star_missing`, elevation, cooling and heating degree days, precipitation, snowfall, snow depth, average temperature, days below 30/20/10/0 °F, days above 80/90/100/110 °F, winter minimum and summer maximum temperature. `Year_Factor` is deliberately excluded: it is an anonymised index, and Year 7 lies outside the training range.

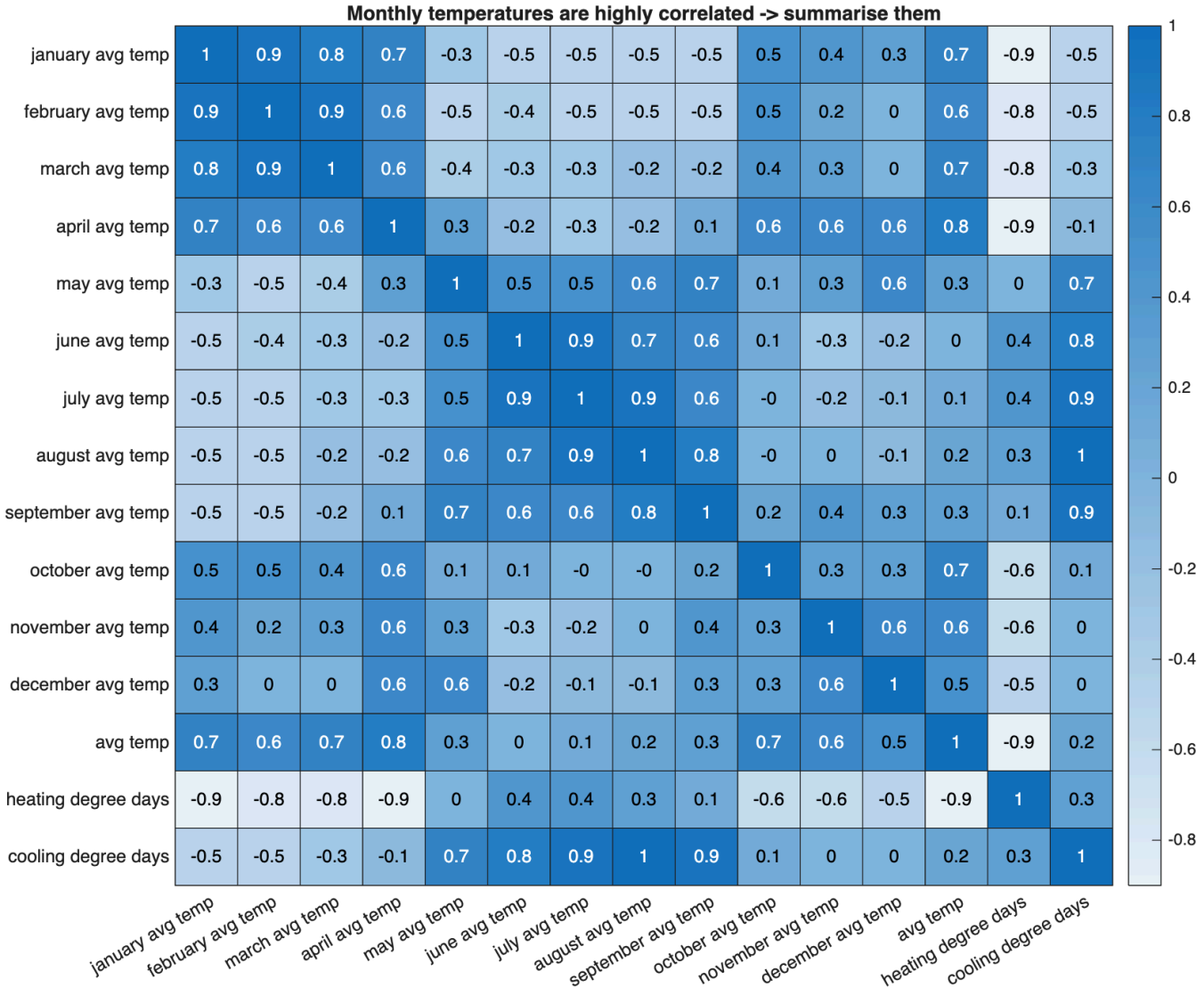


Figure 4. The 12 monthly average temperatures correlate strongly with each other and with degree days, so two seasonal summaries replace the 36 monthly columns.

Step 4: Split by year, not randomly.

Purpose	Train on	Evaluate on
Tune the single tree's leaf size	Years 1–4	Year 5
Tune boosting (time-series cross-validation, fold 1)	Years 1–3	Year 4
Tune boosting (fold 2)	Years 1–4	Year 5
Score every model (final accuracy)	Years 1–5 (53,301 rows)	Year 6 (22,444 rows)
Predict the test year	Years 1–6 (75,745 rows)	Year 7 (9,705 rows, no answers)

Year 6 is never used to make any decision. A random split was rejected because the same building appears in several years: on Year 6, the final model's error is 29.7 for the 8,456 buildings it saw in Years 1–5 and 45.6 for the 13,988 new buildings (figure 11). A random split would let the model recognise buildings rather than predict them, and would overstate accuracy.

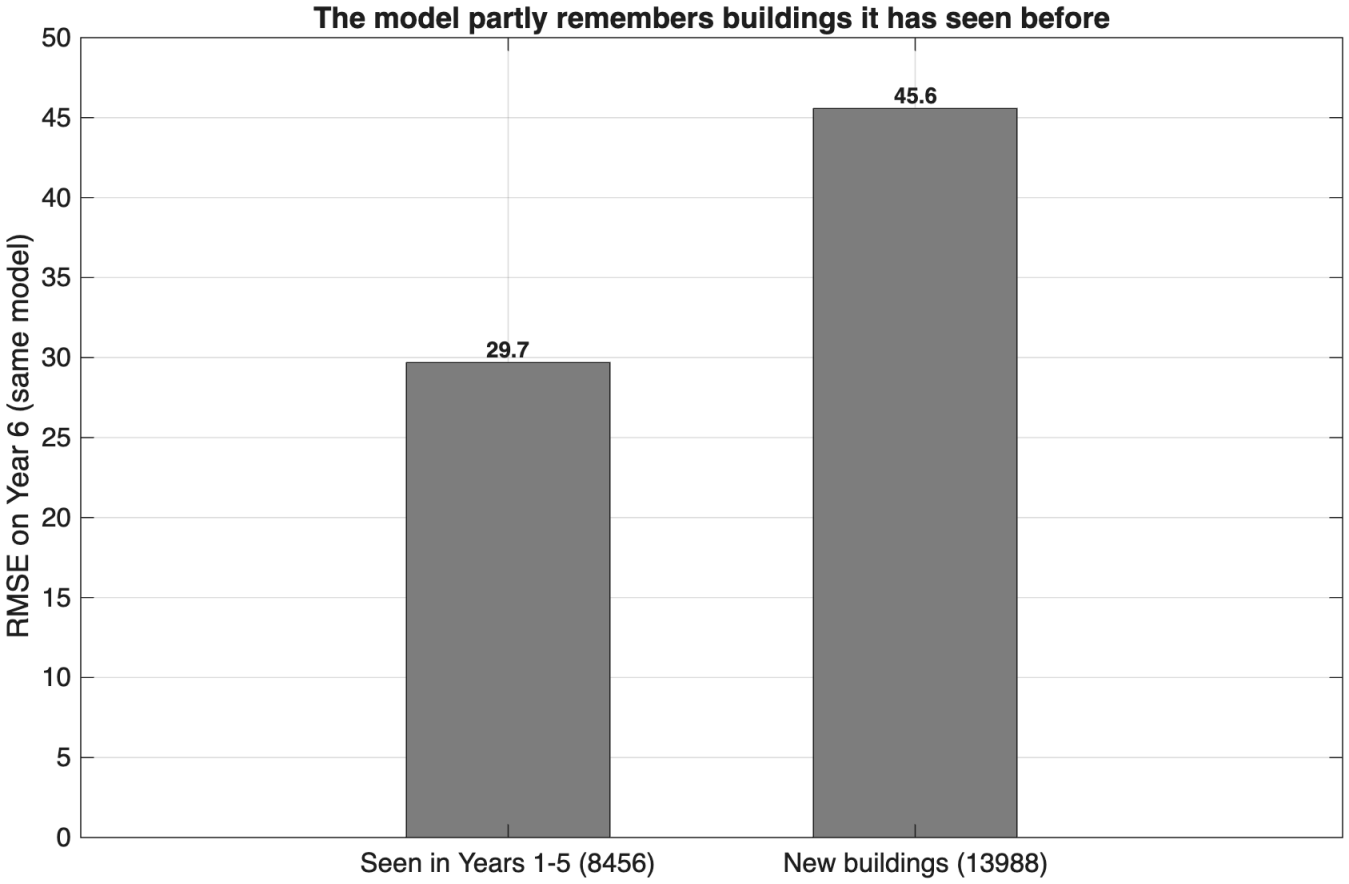


Figure 11. The model is far more accurate on buildings it has seen before, which proves the memorisation risk and justifies the year-based split.

5. Models and results

We built eight models from simplest to most complex. Each is trained on Years 1–5 and scored on Year 6, so every score answers the same question: how well does the model predict a year it has never seen?

#	Model (MATLAB function)	Why it is in the sequence	RMSE	MAE	R^2	RMSE excl. State 6
0	Facility-type median	The bar every real model must clear	50.7	27.2	0.21	53.6
1	Linear regression (`fitlm`)	Uses all inputs, but only straight-line, additive effects	44.6	23.1	0.39	49.1

#	Model (MATLAB function)	Why it is in the sequence	RMSE	MAE	R^2	RMSE excl. State 6
2	Regression tree, leaf size 50 (`fitrtree`)	Captures non-linear effects and interactions	42.7	21.5	0.44	46.8
3	Bagged trees, 100 (`fitrensemble`, Bag)	Averaging many trees reduces variance	42.6	22.7	0.44	44.8
4	Boosted trees, raw target (`fitrensemble`, LSBoost)	Each tree corrects the errors of the previous ones	44.4	22.3	0.39	**40.2**
5	Boosted trees, log target	Same model; log target stops extreme buildings dominating	**40.2**	18.8	**0.50**	42.4
6	Neural network, 64 + 32 neurons (`fitrnet`)	A different model family, to test the tree choice	53.9	29.6	0.11	61.3
7	**Tuned boosted trees, log target (final)**	Tree size, learning rate and tree count chosen by time-series cross-validation	40.3	**18.7**	**0.50**	43.2

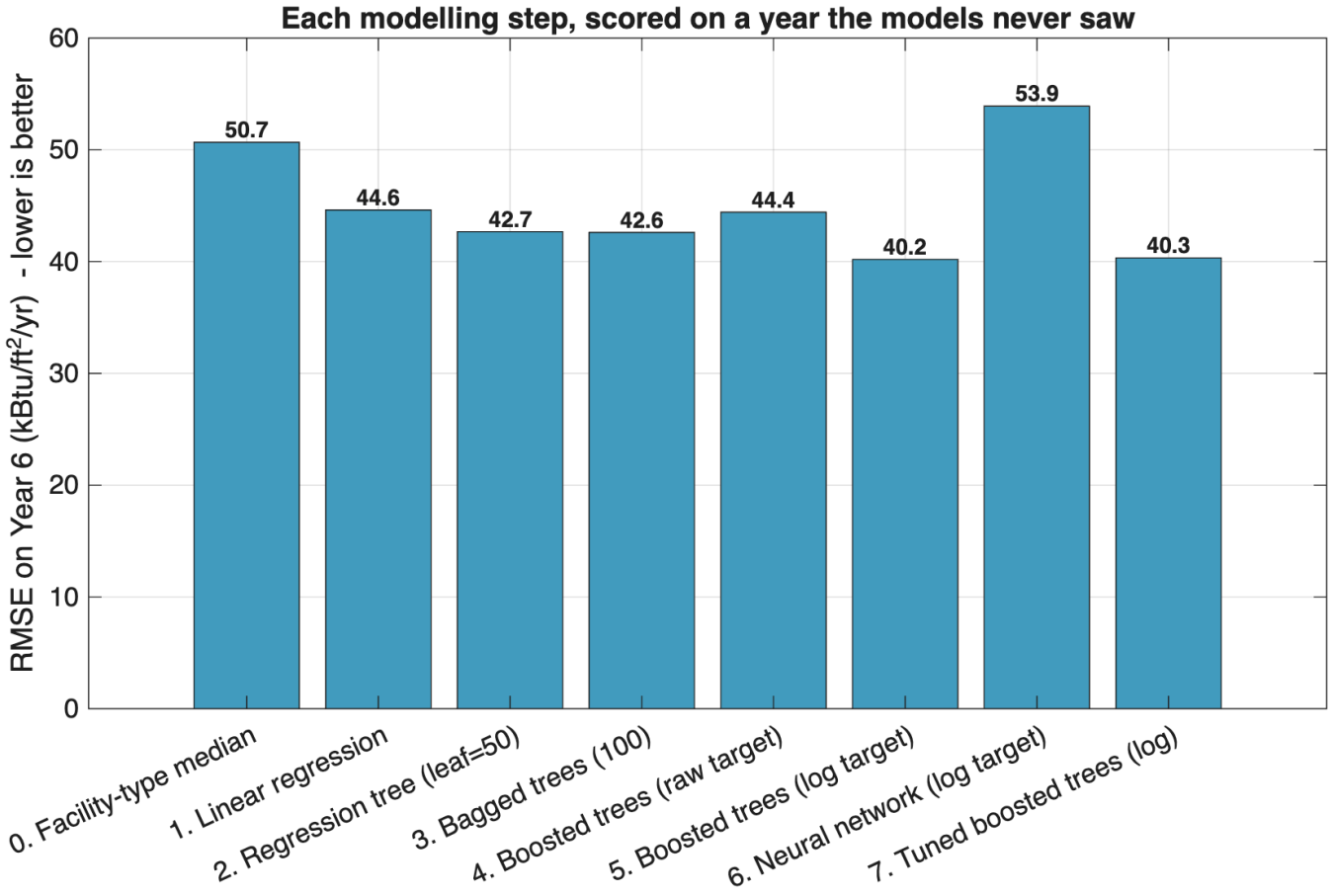


Figure 7. Every model scored on Year 6. The log-target boosted models are the most accurate overall.

What the sequence shows:

1. Using all inputs (linear regression) beats predicting the facility-type median: 50.7 → 44.6.
2. Trees beat linear regression (44.6 → 42.7) because effects interact: the effect of cold weather depends on the building type.
3. Boosting on the raw target (44.4) does worse than bagging, because a few buildings above 900 kBtu/ft²/yr produce huge squared errors that boosting chases.
4. The same boosting model on log(EUI) is the largest single improvement (44.4 → 40.2).
5. The neural network trails every tree model. On tabular data with categorical inputs and heavy outliers, tree ensembles are the established strong choice; the network would need far more tuning.
6. Outside State 6, the raw-target model is best (40.2). State 6 is dominated by multifamily housing, where the log model is strongest. Section 7 explores this.

Tuning, and a lesson from it.

- *Single tree:* leaf sizes 5 to 200 were tried on Year 5; 50 was best (figure 5).
- *Boosting, first attempt:* tuned on Year 5 alone, it chose settings that scored worse on Year 6 (42.8) than the untuned model (40.2). Year 5 contains State 11, which never appears in Years 1–4, so the settings fitted one year's quirks.
- *Boosting, final:* time-series cross-validation over two future years (train 1–3 → check 4, train 1–4 → check 5), with a grid of tree size {15, 30, 60} × learning rate {0.05, 0.1}, including the untuned settings. MATLAB's `loss(model, X, y, 'Mode', 'cumulative')` returns the error after every number of trees from one fit, so the tree count (up to 600) is tuned at no extra cost (figure 6). It chose 298 trees, 60 splits and a learning rate of 0.05, matching the untuned model's accuracy (40.3 vs 40.2) with the lowest MAE, which confirms those settings were already near-optimal.

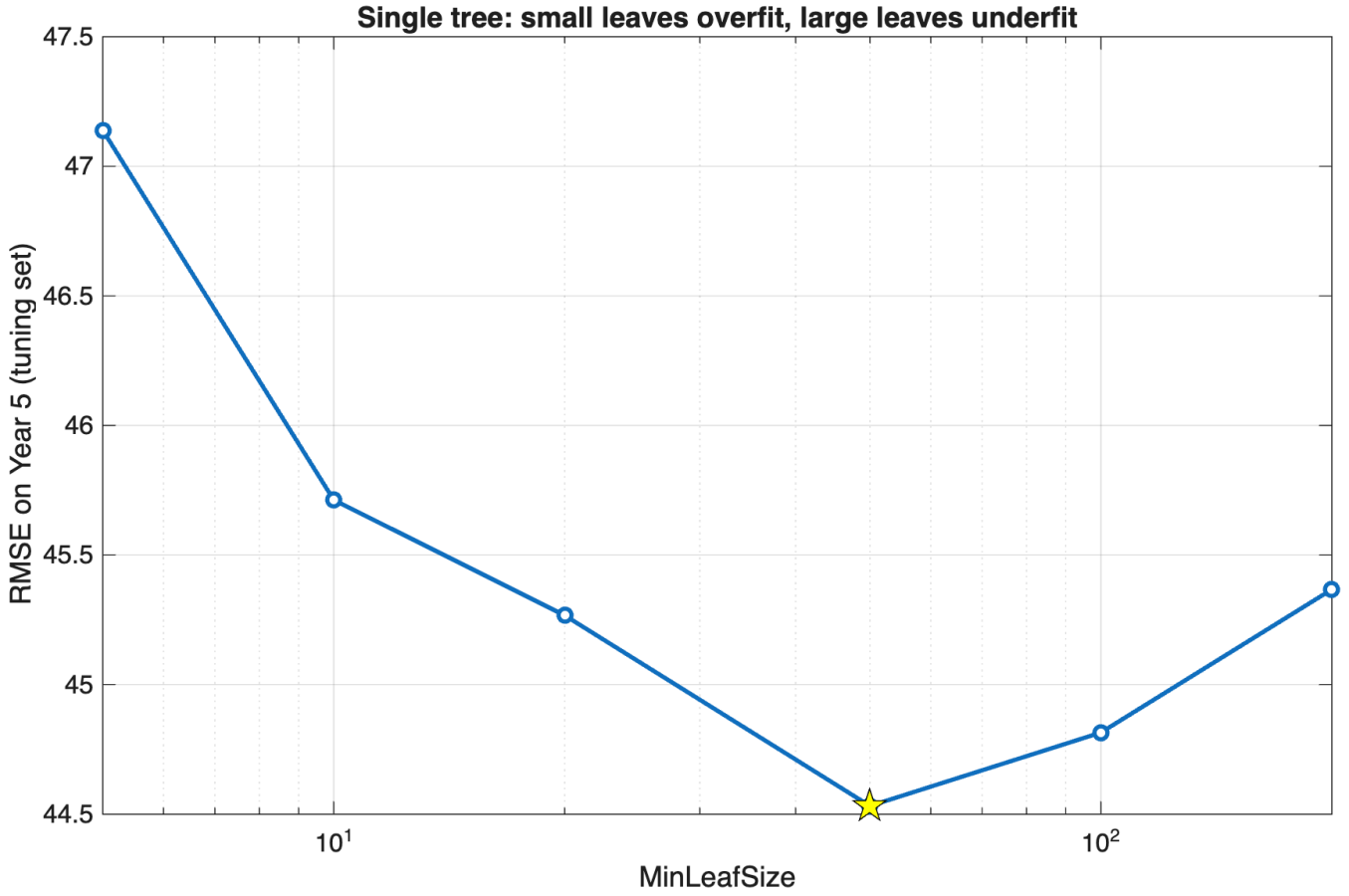


Figure 5. Small leaves overfit and large leaves underfit; leaf size 50 is chosen.

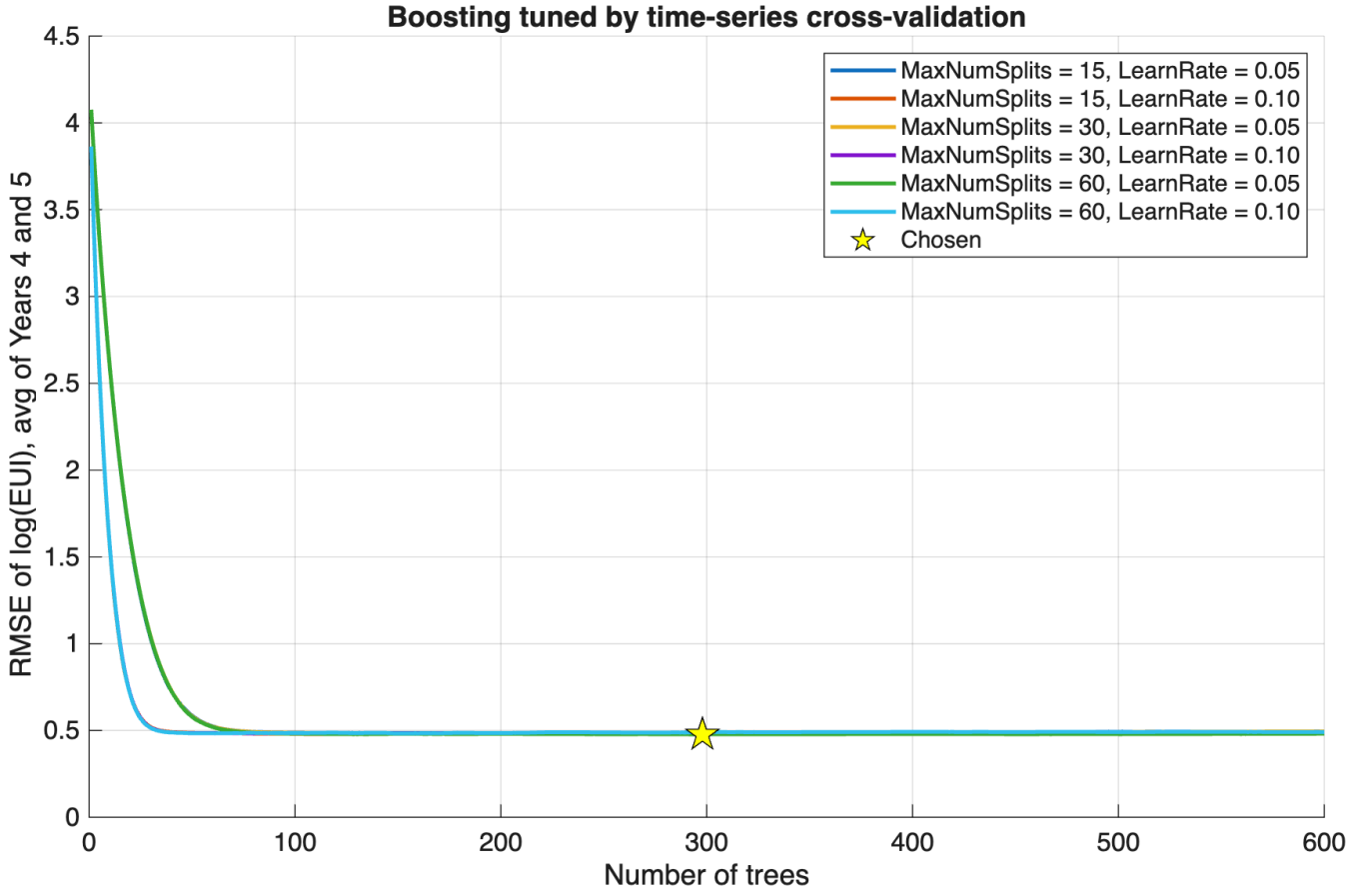


Figure 6. Average error over Years 4 and 5 for each tree size and learning rate; the star marks the chosen setting.

Accuracy and drivers of the final model.

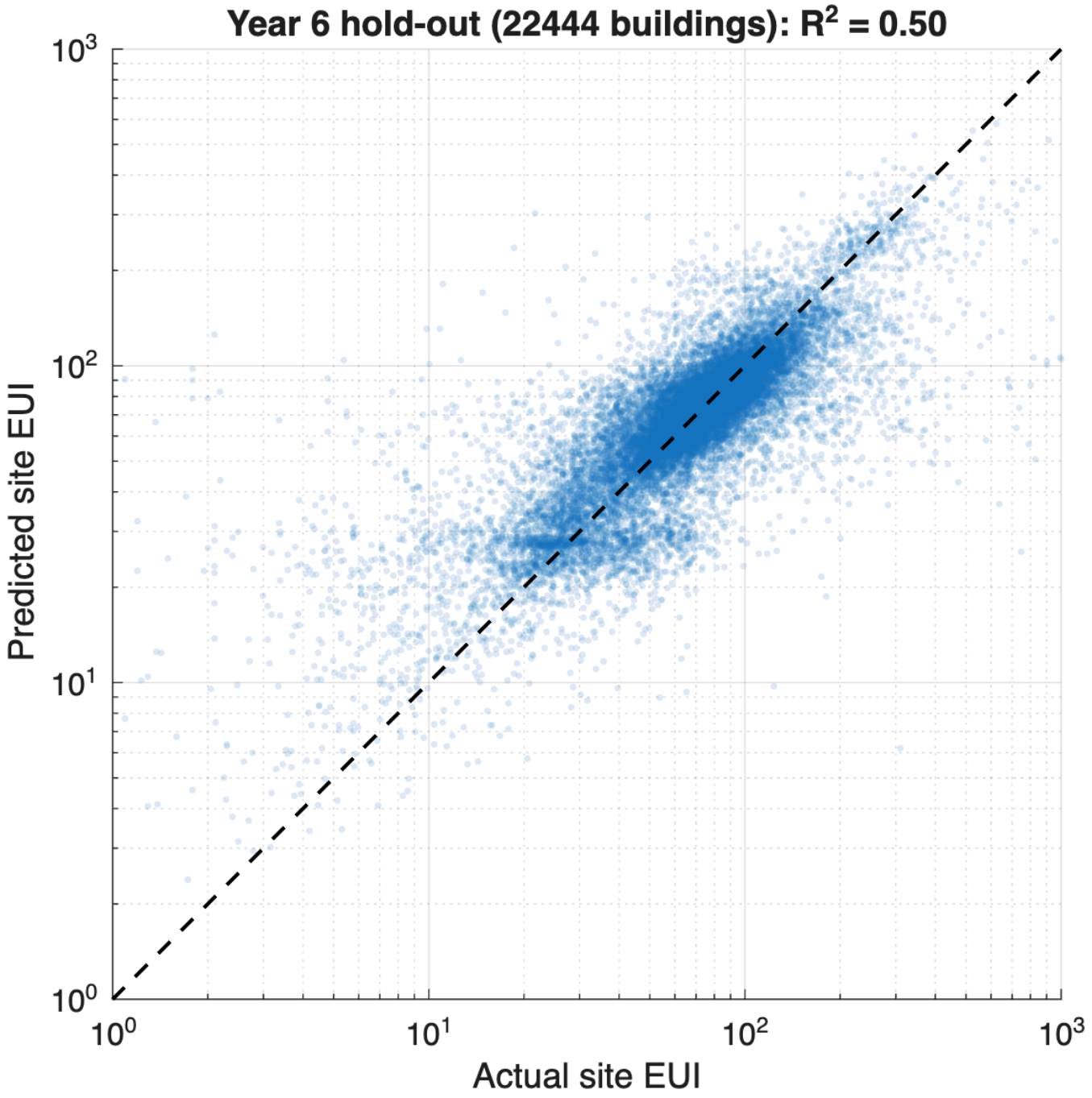


Figure 8. Predicted vs actual on log scales (R^2 0.50). Typical buildings sit near the line; the large misses are buildings with extreme recorded use.

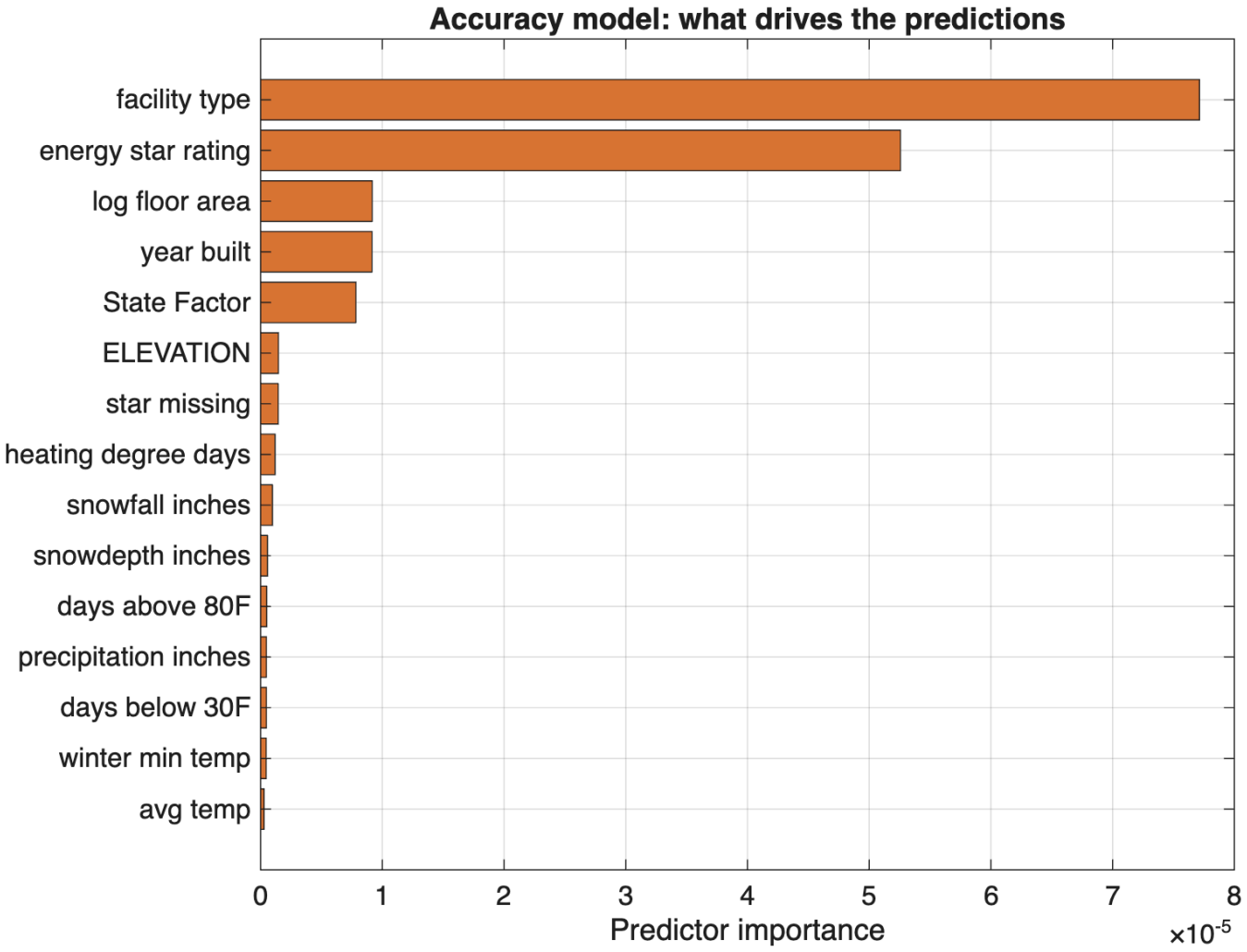


Figure 9. Facility type and Energy Star rating dominate, followed by floor area, year built and state.

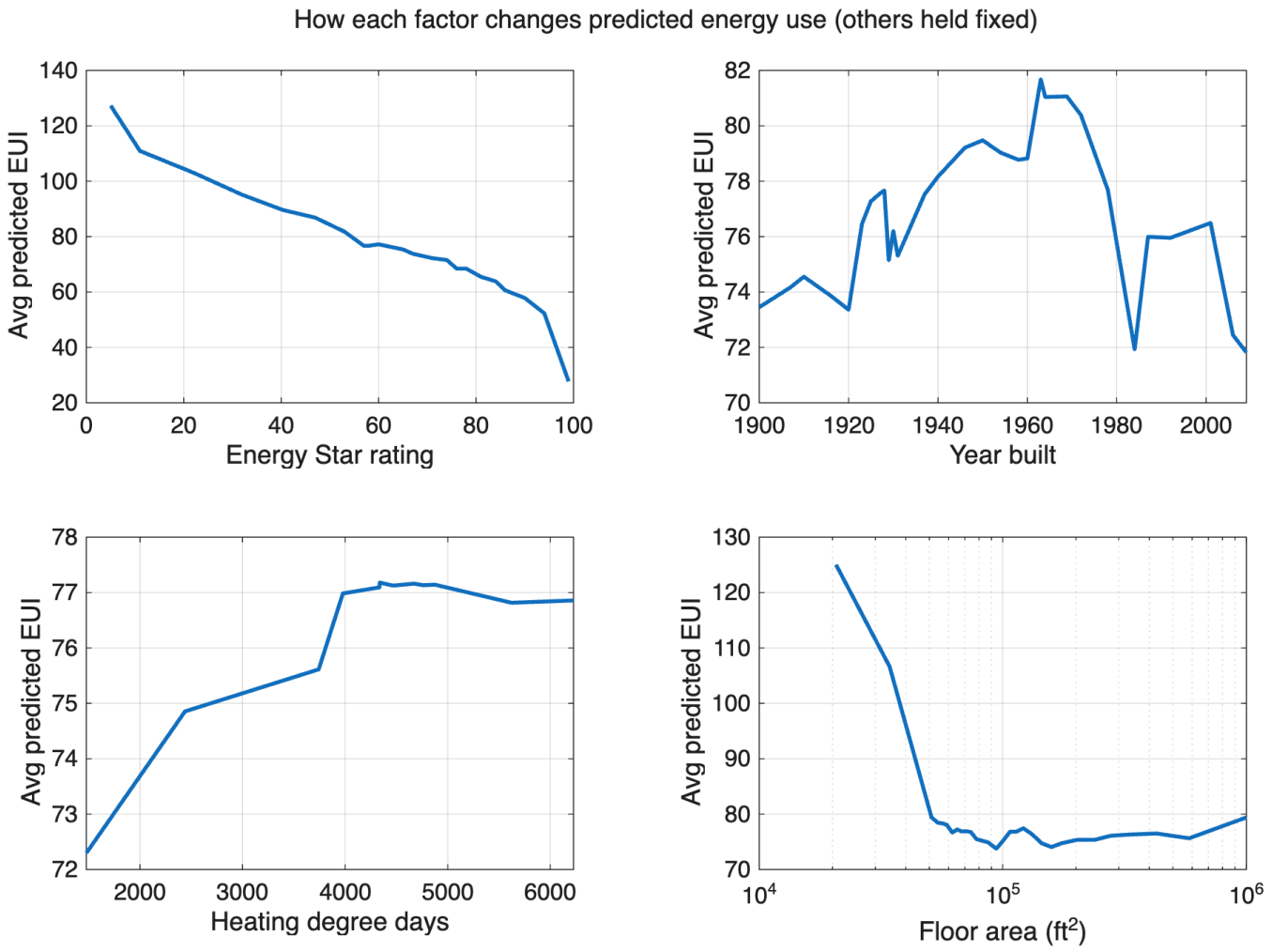


Figure 10. How predicted energy use changes with one input while all others keep their real values.

Input	Effect on predicted energy use (figure 10)
Energy Star rating	Falls steadily as the rating rises: about 127 kBtu/ft²/yr near a rating of 5, about 28 near 100
Year built	Highest for buildings from around the 1960s; lower for buildings built after the 1970s
Heating degree days	Rises with colder climates up to about 4,000 heating degree days, then levels off
Floor area	Smaller buildings use more per square foot; the effect flattens above about 50,000 ft²

These are associations learned from the data, not proof of cause, but they point to where retrofits are likely to matter: low-rated buildings, buildings from the 1960s and 1970s, and buildings in cold climates.

6. The retrofit engine

The retrofit engine turns predictions into a decision: which buildings should be audited for retrofit first?

A fair benchmark model. A second boosted-tree model, with the final model's settings, predicts what each Year 6 building should use. Two rules make the comparison fair:

1. **It never sees the Energy Star rating.** The rating is itself calculated from a building's energy efficiency. A model that saw it would expect poorly rated buildings to use a lot, so they would never look wasteful. Without it, buildings are judged only on type, size, age, state and weather.
2. **It never judges a building it trained on.** It is trained on Years 1–5 with every building that appears in Year 6 removed, using the pseudo building ID. Otherwise it would partly reproduce each building's own past use (section 4, figure 11) and hide buildings that are inefficient every year.

By design it is less accurate than the accuracy model (RMSE 47.7 vs 40.3): it is not supposed to know how efficient a building is.

Calculations for each Year 6 building:

Quantity	Formula
Ratio	actual EUI ÷ expected EUI
Excess EUI	actual EUI − expected EUI (kBtu/ft²/yr)
Excess energy	excess EUI × floor area × 0.000293 (kBtu → MWh)
Excess CO_2	excess energy × 0.38 t/MWh (assumed factor)

Classification:

Outcome	Rule	Buildings	Action
Not flagged	ratio below 1.5	19,364	None
Retrofit candidate	ratio 1.5 to 5	2,973 (13.2%)	Audit, in order of excess energy
Verify meter data	ratio 5 or more	107	Check the meter or the data entry first; not counted as savings

Validation. Because the benchmark never saw Energy Star ratings, the ratings are an independent check. Retrofit candidates have a median rating of **20**; unflagged buildings have **73** (figure 12). Two separate measures of efficiency agree on which buildings are inefficient.

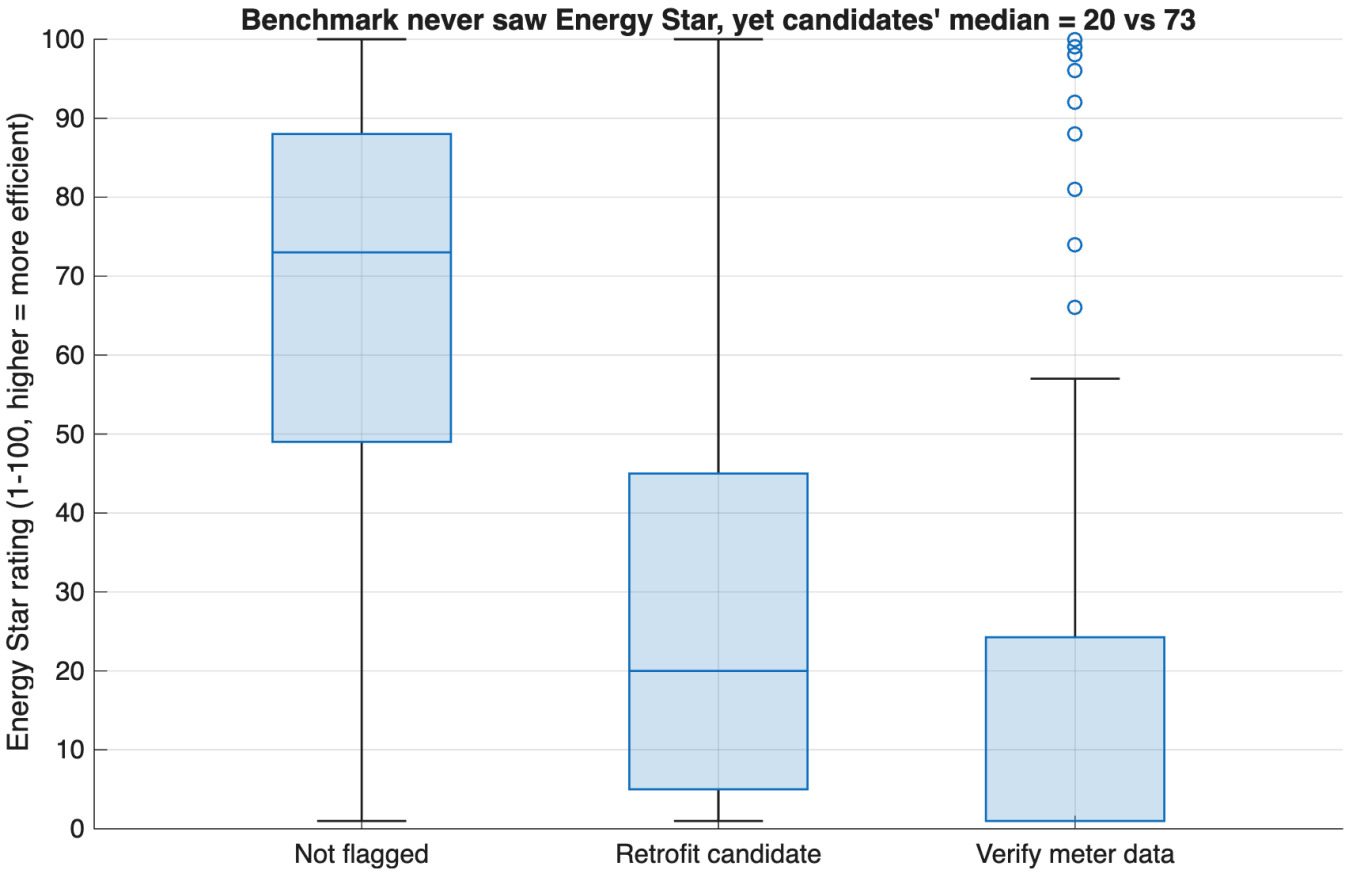


Figure 12. The benchmark never saw Energy Star ratings, yet the buildings it flags score far lower.

Where the waste sits. Ranked by excess energy, the top 5% of buildings hold 57% of all excess energy (figure 13): a limited budget aimed at the top of the list captures most of the opportunity. By building type, the largest totals among candidates are multifamily housing, offices and college or university buildings (figure 14). Together the candidates use about 8,215 GWh a year more than their benchmarks (median about 1,305 MWh per building). This total is an upper bound and is dominated by a few very large buildings.

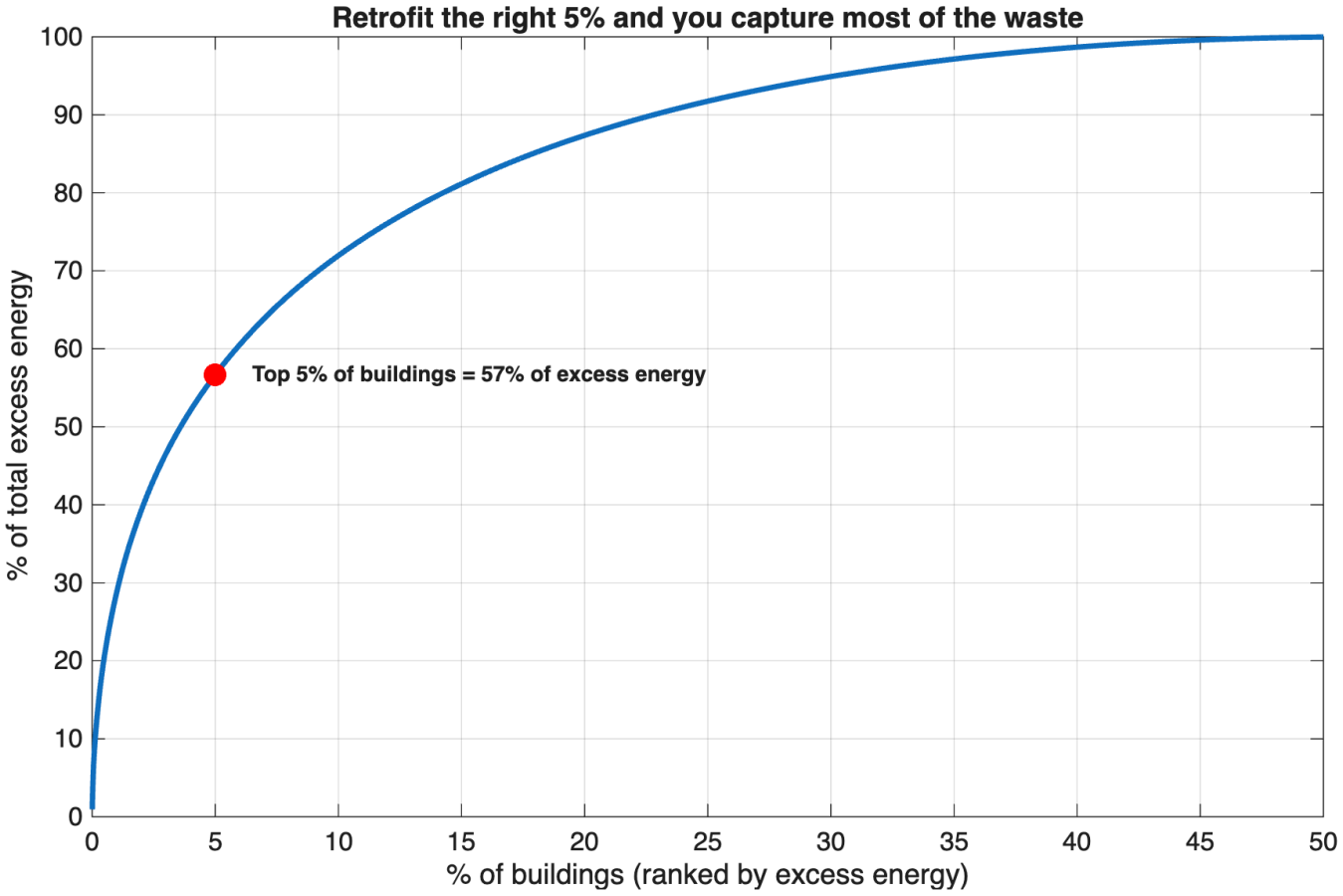


Figure 13. Excess energy is concentrated: the top 5% of buildings hold 57% of it.

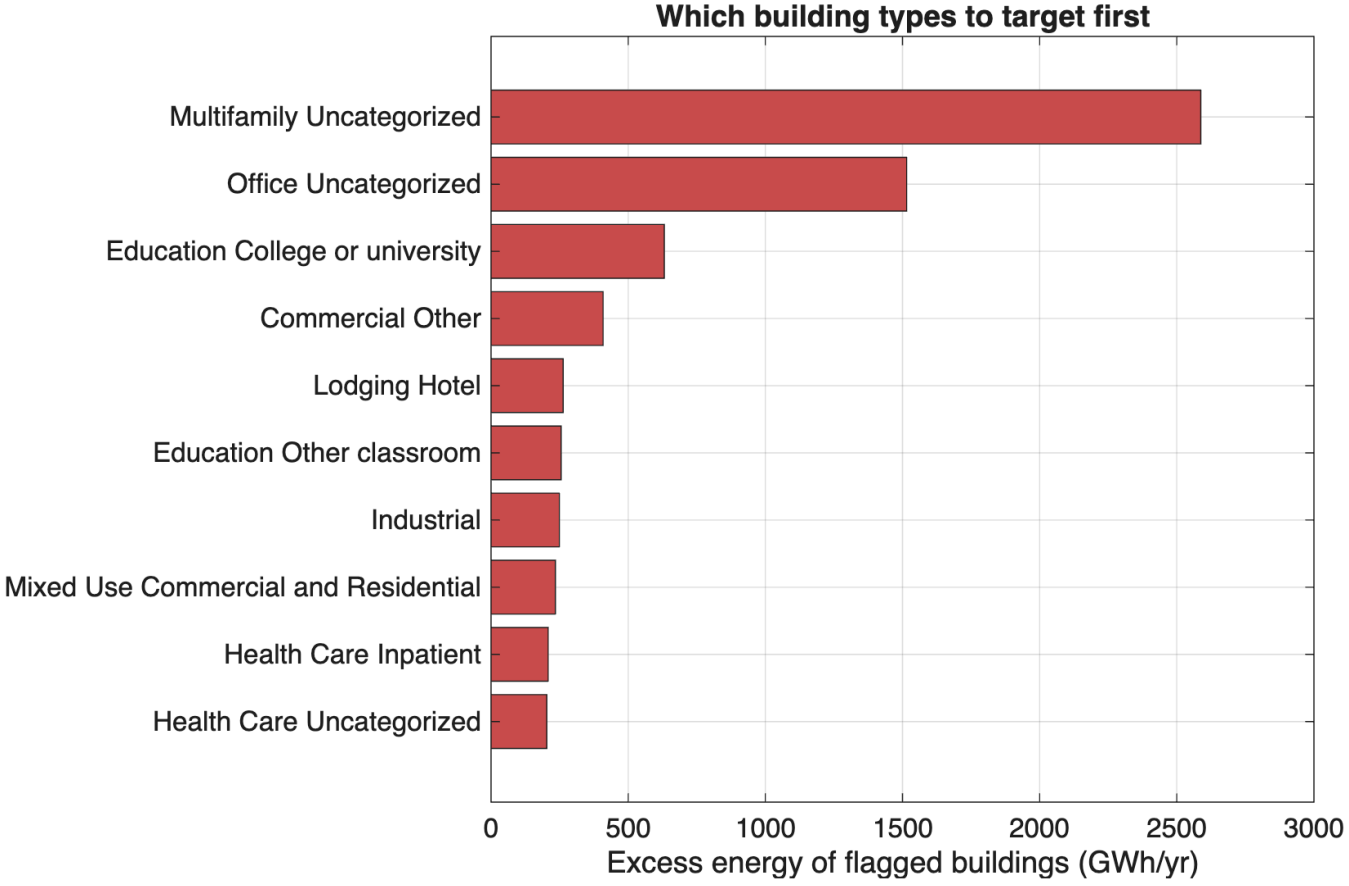


Figure 14. Which building types to audit first.

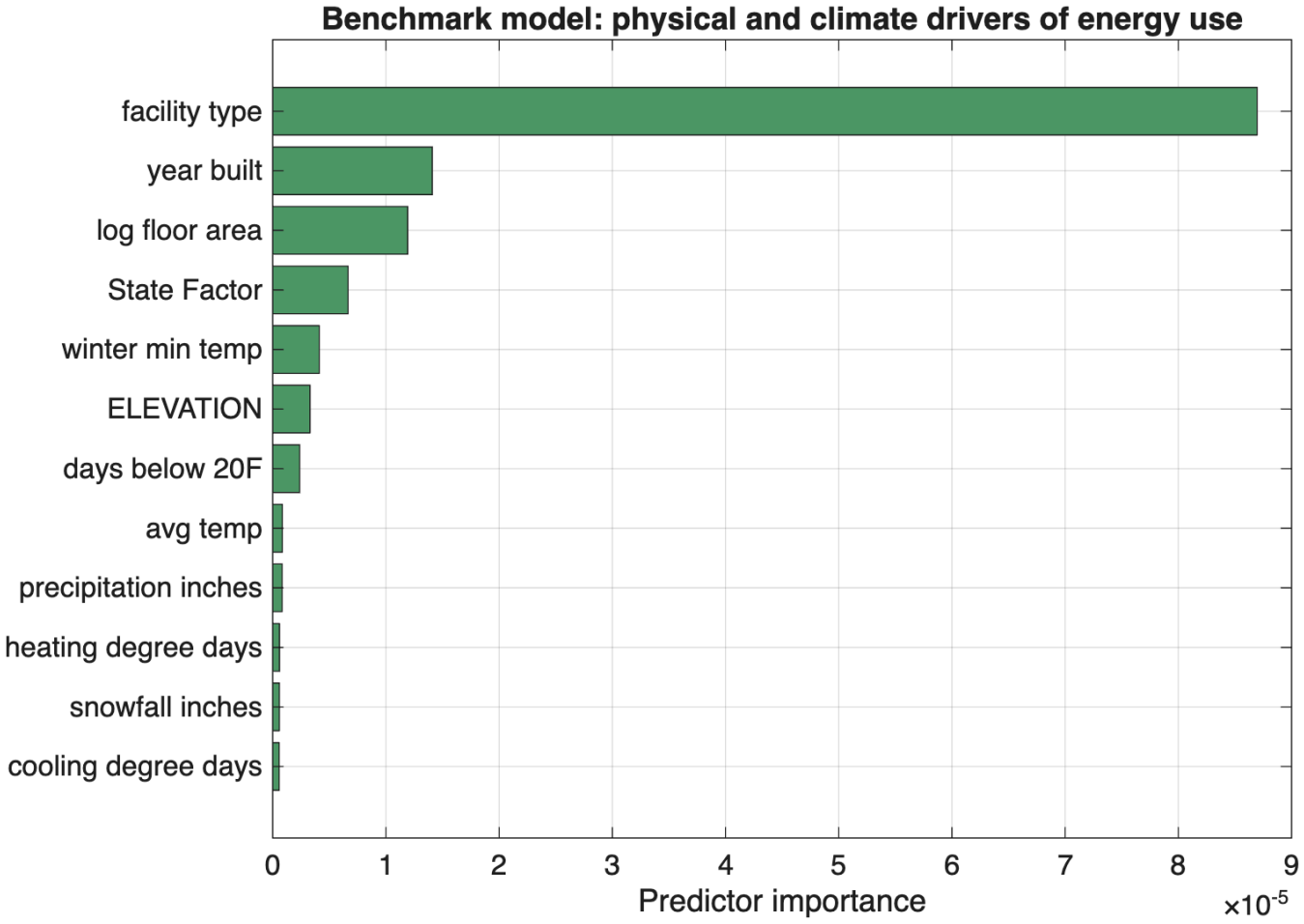


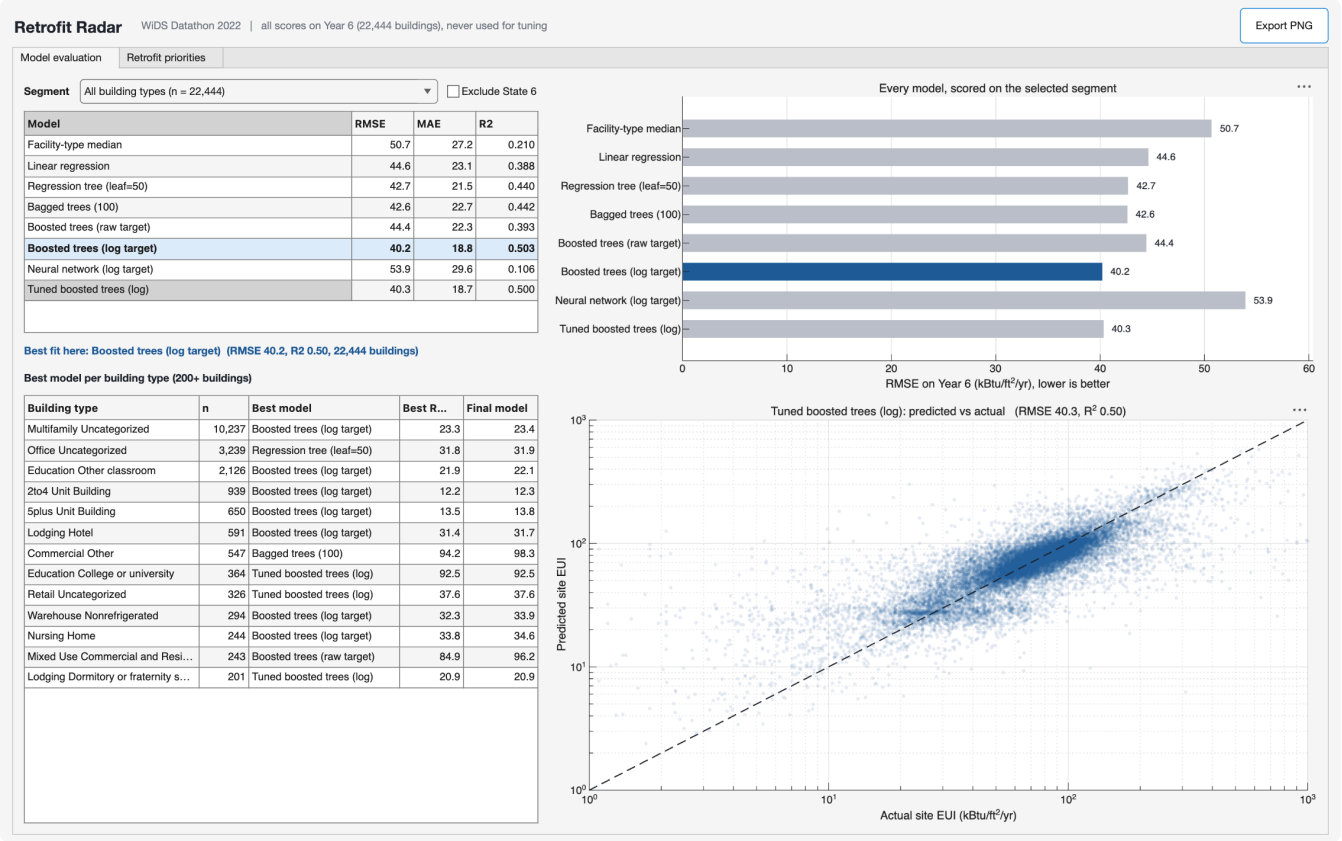
Figure 15. Without the Energy Star rating, the benchmark relies mostly on facility type, then year built, floor area, state and winter temperature.

Outputs: `results/retrofit_priority_list.csv` (one row per candidate, ranked by excess energy, with actual and expected EUI, ratio, excess MWh and tCO_2) and `results/meter_data_checks.csv` (the 107 readings to verify).

7. The MATLAB app

`RetrofitRadarApp.m` is an interactive app built entirely in code (`uifigure`, `uigridlayout`, `uitabgroup`, `uitable` with `uistyle` highlighting, `uiaxes`, `uislider`, nested callbacks). It reads the saved results, so it opens in seconds without retraining. It has two tabs.

Model evaluation. All eight models' Year 6 predictions are saved by the pipeline (`results/year6_model_predictions.csv`). The app re-scores every model on any building type, with or without State 6, highlights the best fit, and lists the best model for each building type with 200 or more buildings. Clicking a model shows its predicted-vs-actual plot.



App, all Year 6 buildings: log-target boosting is best (RMSE 40.2), with the final tuned model at 40.3.

Image — App: model evaluation, State 6 excluded

App, State 6 excluded: raw-target boosting becomes best (RMSE 40.2 vs 43.2 for the final model).

What the app revealed. The best model depends on which buildings are scored.

Segment	Best model	Best RMSE	Final model RMSE
All Year 6 buildings	Boosted trees, log target	40.2	40.3
Excluding State 6	Boosted trees, raw target	40.2	43.2
Multifamily (all states, 10,237 buildings)	Boosted trees, log target	23.3	23.4
Office (3,239)	Regression tree	31.8	31.9
Commercial other (547)	Bagged trees	94.2	98.3

Segment	Best model	Best RMSE	Final model RMSE
College or university (364)	Tuned boosted trees	92.5	92.5
Mixed use (243)	Boosted trees, raw target	84.9	96.2

State 6 holds 80% of the Year 6 multifamily buildings, which is where the log model is strongest, so it wins overall but not outside State 6. We kept the final model chosen by our pre-set rule (time-series cross-validation) rather than switching after seeing Year 6. For the Year 7 buildings, which include no State 6, a raw-target model or per-type model selection is the clearest next step.

Retrofit priorities. The ranked candidates (or the meter-data checks), filtered by building type and a minimum actual/expected ratio from 1.5 to 5. A summary line updates the count, excess energy, CO_2 and median Energy Star rating. Clicking a building shows its actual vs expected energy and its details; a concentration curve shows how much of the list's excess sits in its top 10%.

Image — App: retrofit priorities

App, retrofit priorities: the top-ranked building uses 1.8 times what comparable buildings use (342 vs 192 kBtu/ft²/yr).

8. Assumptions

Every assumption we made, why we made it, and what changes if it is wrong.

Data and cleaning

#	Assumption	Why	If it is wrong
A1	Year 7 behaves like a future year, so validating on a later year (Year 6) mimics the real task	`test.csv` contains only Year 7	Our Year 6 scores would misstate Year 7 accuracy; the State 6 shift is the main known difference, which is why we also report RMSE excluding State 6

#	Assumption	Why	If it is wrong
A2	Rows with identical state, facility type, floor area, year built and elevation are the same building	No building ID is provided; these attributes rarely coincide by chance	Two genuinely different buildings could be merged, or a building whose floor area changed could be split; the memorisation measure would shift slightly
A3	Columns more than 50% empty carry too little information to keep	Wind and fog are 54–60% empty in training and 88–94% in the test year	We might lose a small wind effect; the remaining weather inputs cover temperature, snow and degree days
A4	Missing Energy Star ratings and build years are best estimated by the median of the same facility type	Type is the strongest single driver of both	If rated buildings differ systematically from unrated ones, fills are biased; the `star_missing` flag lets the model adjust for this
A5	Build years before 1800 are data-entry errors	Values such as 0 are impossible	Negligible: 25 rows
A6	Exact duplicate rows are errors, not separate buildings	Every column, including the target, matches	Negligible: 12 rows
A7	Winter minimum and summer maximum temperatures capture the useful monthly temperature information	The 36 monthly columns are highly correlated with each other and with degree days	Some seasonal detail is lost; degree days and the days-below/above columns remain as inputs
A8	The weather recorded for a building's location and year represents the conditions it experienced	Weather is given per location and year	Local microclimates are not captured

Modelling

#	Assumption	Why	If it is wrong
A9	`Year_Factor` should not be a model input	It is an anonymised index, and Year 7 lies outside the training range	Any genuine year-on-year trend in energy use is not modelled
A10	Modelling log(EUI) suits the data	The target is heavily right-skewed	Back-transforming with exp() estimates something closer to the median than the mean, so predictions for very high users run low; we did not apply a bias correction
A11	Tuning on earlier years (time-series cross-validation over Years 4 and 5) picks settings that transfer to later years	It mirrors predicting the future	Settings may be sub-optimal for the test year; outside State 6 a raw-target model already does better
A12	Our hyperparameter grid (tree size 15/30/60, learning rate 0.05/0.1, up to 600 trees) is wide enough	Limited hackathon time; the number of trees is searched exhaustively via cumulative loss	A wider search might gain a little accuracy
A13	One fixed random seed (42) gives representative results	Reproducibility	Bagging and the neural network vary slightly with other seeds
A14	The neural network's settings (64 + 32 neurons, standardised inputs, light regularisation, 500 iterations) are reasonable defaults	Included as a comparison, not tuned	A tuned network could do better than reported, though trees are the established choice for this kind of data

Retrofit engine

#	Assumption	Why	If it is wrong
A15	“Comparable building” means the same type, size, age, state and weather	These are the physical and climate facts available	Unrecorded factors (occupancy, opening hours, equipment) can make an efficient building look wasteful
A16	The Energy Star rating measures efficiency, so the benchmark must not see it, and it can serve as an independent check	The rating is calculated from energy use relative to peers	If the rating were unrelated to efficiency, the 20 vs 73 validation would mean less
A17	A building using 1.5× or more of its expected energy is worth auditing	A practical threshold that flags about 13% of buildings	A policy choice: lower flags more, higher flags fewer (adjustable in `cfg.flagRatio` and in the app)
A18	A reading of 5× or more of expected energy is more likely a meter or data error than a real building	Such values are physically implausible for most building types	Some real extreme users are set aside for checking instead of being counted (107 buildings)
A19	Excess energy is what would be saved if a building matched its benchmark	It measures the gap to comparable buildings	Real savings depend on which retrofits are feasible; the totals are an upper bound
A20	0.38 tonnes of CO_2 per MWh	Approximate US grid electricity average	Site energy also includes gas and other fuels with different factors; the CO_2 total is indicative only
A21	1 kBtu = 0.000293071 MWh	Physical unit conversion (not an assumption about the data)	—

App

#	Assumption	Why	If it is wrong
A22	Segments need at least 200 buildings (50 after excluding State 6) to compare models	Smaller groups give unstable error estimates	Small building types are not shown in the per-type table

9. Limitations and next steps

Limitations

- **A screening tool, not an audit.** A flag says a building uses much more than comparable buildings; it does not say why or what to fix.
- **Missing drivers cap accuracy.** The data has no occupancy, opening hours, equipment or maintenance information, so about half the variation between buildings (R^2 0.50) is unexplained.
- **Indicative CO_2.** The emission factor is a single assumed average (A20).
- **Upper-bound savings.** Excess energy assumes each candidate could reach its benchmark exactly (A19), and the total is dominated by a few very large buildings.
- **The final model is not best everywhere.** Outside State 6, a raw-target model is more accurate (section 7).
- **Anonymised data.** Years and states are masked, so results cannot be tied to real places or dates.

Next steps

1. Choose models per building type, or switch to the raw-target model for Year 7 (no State 6), and validate the choice on a held-out year.
2. Add retrofit cost estimates per building type to rank buildings by cost per tonne of CO_2 saved.
3. Re-run every year as new meter data arrives, and track whether flagged buildings improve after retrofit.
4. Add occupancy and operating-hours data where available to raise accuracy.
5. Apply a bias correction when converting log predictions back to energy use (A10).

10. Glossary

Energy and buildings

Term	Definition
Site EUI (energy use intensity)	Energy a building uses on site in a year divided by its floor area, in kBtu per square foot per year. The target we predict.
kBtu	Thousand British thermal units, a US energy unit. 1 kBtu = 0.293 kWh = 0.000293 MWh.
MWh / GWh	Megawatt-hour / gigawatt-hour of energy. 1 GWh = 1,000 MWh.
kt CO_2	Thousand tonnes of carbon dioxide.
Emission factor	Tonnes of CO_2 released per MWh of energy used. We assume 0.38 t/MWh.
Building-year	One row of data: one building in one year.
Facility type	The building's use (for example office, multifamily housing, hospital, warehouse); 60 types in the data.
Building class	Residential or commercial.
Energy Star rating	A 1–100 score from the US Environmental Protection Agency comparing a building's energy efficiency with similar buildings; higher means more efficient.
Heating / cooling degree days	Measures of how much heating or cooling a climate demands over a year: the sum over all days of how far the daily temperature sits below (heating) or above (cooling) a baseline comfort temperature.
Retrofit	Upgrading an existing building to use less energy (insulation, heat pumps, HVAC, glazing, lighting).
Retrofit candidate	A building using 1.5 to 5 times its expected energy.
Verify meter data	A building using 5 or more times its expected energy, treated as a likely meter or data error.
Excess energy	(Actual EUI − expected EUI) × floor area, converted to MWh per year.

Data handling

Term	Definition
Categorical variable	A label rather than a number (state, facility type); stored as MATLAB `categorical`.
Missing value / imputation	An empty cell / filling it in with an estimate (we use the facility-type median).
Distribution shift	When the data a model is tested on differs from the data it was trained on (here, State 6 is in training but not in the test year).
Data leakage	Information from the test data reaching the model during training, which makes accuracy look better than it really is.
Memorisation	A model predicting a building well because it has seen that building before, rather than because it learned general patterns.
Pseudo building ID	Our grouping of rows with the same state, facility type, floor area, year built and elevation, treated as one building across years.
Log transform	Modelling log(EUI) instead of EUI to compress the long right tail; predictions are converted back with exp().
Correlation	How strongly two variables move together, from −1 to 1.

Modelling and evaluation

Term	Definition
Training / tuning / test set	Data used to fit a model / to choose its settings / to report its final score (used once, for no decisions).
Hold-out year	A year kept out of training and used only for evaluation (Year 6).
Hyperparameter	A setting chosen before training, such as tree size, learning rate or number of trees.
Cross-validation	Scoring settings on several held-out portions and averaging, so the choice does not depend on one portion.

Term	Definition
Time-series cross-validation	Cross-validation where each fold trains on earlier years and validates on a later year.
Baseline	A deliberately simple model (here, the median EUI of each facility type) that every real model must beat.
Linear regression	Predicts the target as a weighted sum of the inputs.
Regression tree	Splits the data with yes/no questions and predicts the average of each final group (leaf). Leaf size is the minimum number of rows per group.
Bagging (bagged trees)	Training many deep trees on random resamples of the data and averaging them.
Gradient boosting (LSBoost)	Adding small trees one at a time, each fitted to the errors left by the trees before it. The learning rate scales how much each tree corrects.
Neural network	Layers of weighted sums with non-linear activations, trained by gradient-based optimisation.
Cumulative loss	MATLAB's ensemble error after 1, 2, … N trees, returned from a single trained model.
RMSE	Root mean squared error: the typical size of a prediction error, with large errors weighted heavily. Lower is better.
MAE	Mean absolute error: the average size of a prediction error. Lower is better.
R^2	The share of variation between buildings that the model explains (0 = no better than the average, 1 = perfect).
Predictor importance	How much each input contributes to a tree ensemble's predictions.
Partial dependence	How the average prediction changes as one input varies while the others keep their real values.
Benchmark model	Our second model, which predicts what a building should use from physical and climate facts only, with no Energy Star rating.

Term	Definition
Ratio (actual / expected)	A building's actual EUI divided by the benchmark's expected EUI.
Pareto (concentration) curve	Cumulative share of the total excess energy against the share of buildings, sorted from largest excess down.
Segment	A subset of buildings, such as one facility type, used to re-score the models in the app.

11. Reproducing our work

Repository: github.com/chuaguanyu0718-dotcom/building-energy-prediction

Requirements: MATLAB R2023a or later (developed on R2026b) with the Statistics and Machine Learning Toolbox.

How to run:

1. Open MATLAB in the repository folder. The data (`train.csv`, `test.csv`) is already in `data/`.
2. Run `main` for the full pipeline (about 12 minutes). Set `cfg.fastMode = true` in `BuildingEnergyRetrofit.m` for a 3-minute check with fewer trees.
3. Run `RetrofitRadarApp` to open the interactive app.

Results are reproducible: the random seed is fixed (`rng(42)`), and repeated runs gave identical numbers.

File	Contents
`main.m`	Entry point; runs the pipeline
`BuildingEnergyRetrofit.m`	The full pipeline in numbered sections (opens as a Live Script)
`RetrofitRadarApp.m`	The interactive app
`data/train.csv`, `data/test.csv`	WiDS Datathon 2022 data
`figures/`	All 15 figures in this report, plus app screenshots
`results/model_comparison.csv`	Year 6 RMSE, MAE, R^2 and RMSE excluding State 6 for all 8 models

File	Contents
`results/year6_model_predictions.csv`	Every model's Year 6 prediction for every building
`results/retrofit_priority_list.csv`	The 2,973 ranked retrofit candidates
`results/meter_data_checks.csv`	The 107 readings to verify
`results/year7_predictions.csv`	Predicted site EUI for the 9,705 unlabelled Year 7 buildings
`results/summary.txt`	Headline numbers
`results/models.mat`	The final accuracy model and benchmark model (compact)